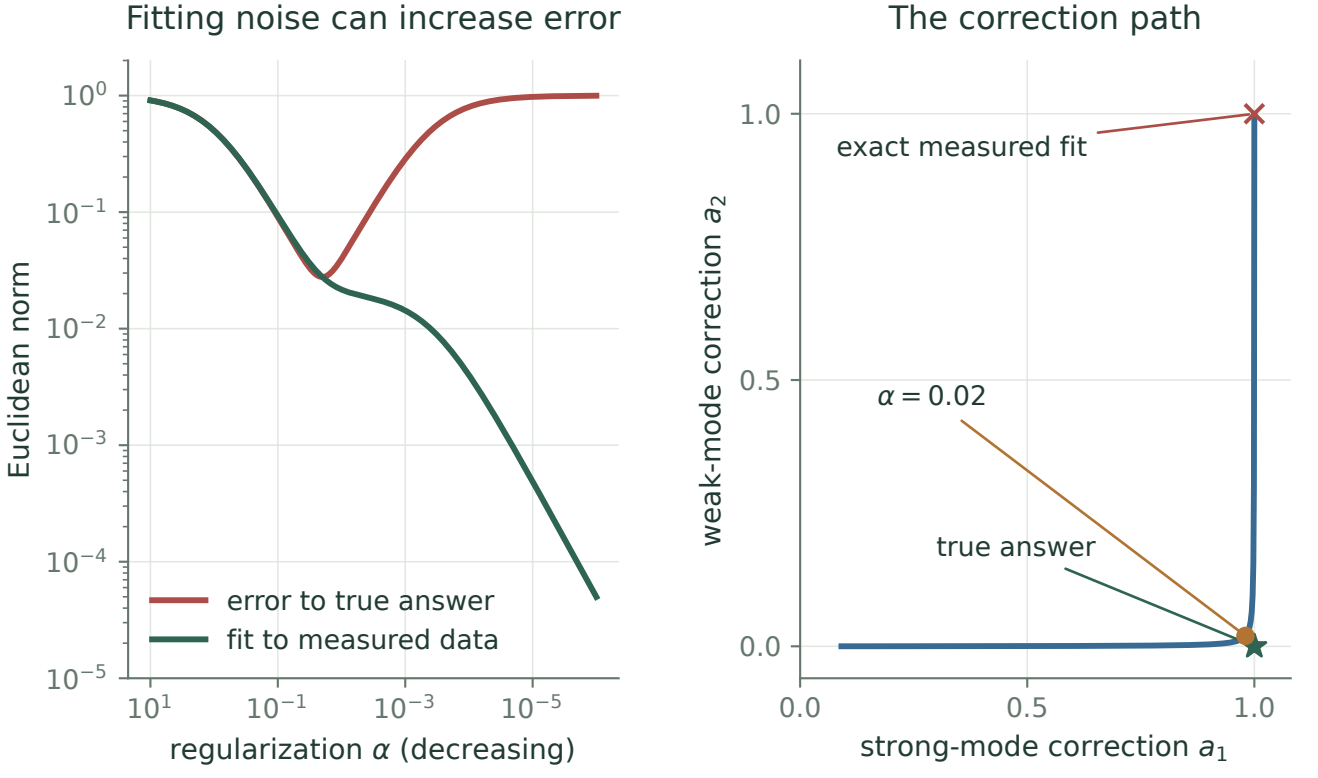
Fitting noise can increase error
Euclidean norm
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
error to true answer
fit to measured data
10^{1}
10^{-1}
10^{-3}
10^{-5}
regularization α (decreasing)
The correction path
weak-mode correction a_2
1.0
0.5
0.0
exact measured fit
$\alpha = 0.02$
true answer
0.0
0.5
1.0
strong-mode correction a_1